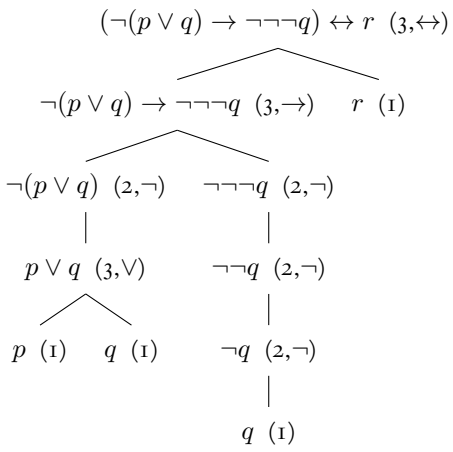


Figure 3.1: Construction tree $(\neg(p \vee q) \to \neg\neg\neg q) \leftrightarrow r$

b. Give all formulas that can be made out of the following sequence of symbols by supplying brackets: $p \wedge q \to r$. Also supply their construction trees.

c. Classify each of the following sentences as an atomic formula, a negation, a conjunction, a disjunction, an implication, or an equivalence.

(i.) $p \to q$
(ii.) $\neg p$
(iii.) p
(iv.) $(p \wedge q) \wedge (q \wedge p)$
(v.) $\neg(p \to q)$
(vi.) $(p \to q) \vee (q \to \neg\neg p)$
(vii.) $\neg(p_1 \wedge p_2) \wedge \neg p_2$
(viii.) $(p \wedge (q \wedge r)) \vee p$

Construction trees do not only provide a nice way to visualise the syntactic structure of a formula, they are also useful for proving things. For that we take another look at last clause of Definition 1, which reads: